

# HYPERLIPIDEMIA: OVERVIEW

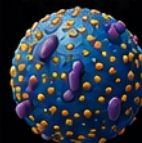
Excess circulating lipids that silently drive atherosclerosis—the modifiable engine behind most heart attacks and strokes.

## WHAT IT IS

- Elevated plasma lipids—chiefly **LDL-C**, but also triglycerides and **lipoprotein remnants**—due to overproduction or impaired clearance.
- Primary (genetic) causes include **familial hypercholesterolemia** (LDL-receptor defect). Secondary causes include diet, obesity, diabetes, hypothyroidism, nephrotic syndrome, cholestasis, and drugs.
- Retained **ApoB-containing** particles enter the arterial intima, oxidize, and trigger foam-cell/plaque formation—the basis of atherosclerotic cardiovascular disease (**ASCVD**).
- Triglycerides  $\geq 500$  mg/dL increase pancreatitis risk;  $\geq 1000$  mg/dL require urgent pancreatitis-prevention management.

## UNDERSTANDING BLOOD LIPIDS

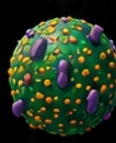
### CHYLOMICRON



Largest  
Carries dietary triglycerides.

ApoB-48

### VLDL



Carries endogenous triglycerides to tissues.

ApoB-100

### LDL

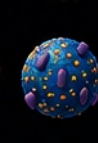


Transports cholesterol to cells; excess LDL may be retained in artery wall.

ApoB-100

"Bad cholesterol"

### HDL



Returns cholesterol to the liver.

ApoA-I

"Good cholesterol"

ApoB (Apolipoprotein B) = Atherogenic protein  
Cholesterol  
Triglyceride

## ★ NEW: Lp(a)

Measure Lp(a) at least once in every adult.

It is genetically determined and not included in a standard lipid panel. Elevated Lp(a) increases ASCVD risk independent of LDL-C.



ApoB counts **atherogenic particles**—not merely the cholesterol carried inside them.

## THE PATH FROM EXCESS LIPIDS TO HEART ATTACK OR STROKE

### 1 NORMAL ARTERY



Healthy endothelium maintains normal blood flow.

### 2 ENDOTHELIAL ACTIVATION



Risk factors cause endothelial activation and increased permeability.

### 3 LIPID ENTRY & OXIDATION



LDL particles (ApoB) enter intima and become oxidized, triggering inflammation.

### 4 FOAM CELL & FATTY STREAK



Macrophages ingest oxidized LDL and become foam cells, forming a fatty streak.

### 5 PLAQUE DEVELOPMENT



Smooth muscle cell proliferation and extracellular matrix create an atherosclerotic plaque.

### 6 PLAQUE RUPTURE & THROMBOSIS



Rupture/erosion exposes thrombogenic material → platelet activation → thrombosis.

### 7 CLINICAL EVENT



Myocardial infarction or ischemic stroke from reduced or blocked blood flow.

## LIPID FRACTIONS THAT MATTER

- LDL-C** ↑ Drives atherosclerosis
- HDL-C** — Lower levels may indicate higher risk; raising HDL-C itself is not a primary target
- Triglycerides** ↑ Energy storage; very high levels increase pancreatitis risk

## WHY IT MATTERS



Cardiovascular disease remains the leading global cause of death, and atherosclerosis accounts for much of its burden.



Most ASCVD risk is influenced by cumulative exposure to atherogenic lipoproteins plus modifiable factors.



Lowering LDL-C with lifestyle and medications dramatically reduces heart attacks, strokes, and death.

## LIPID VALUES THAT REQUIRE ATTENTION

### LDL-C

- LDL-C  $\geq 190$  mg/dL: severe hypercholesterolemia; evaluate for familial hypercholesterolemia
- LDL-C  $\geq 160$  mg/dL in young adults; consider earlier treatment, especially with premature ASCVD family history
- Treatment goals depend on total ASCVD risk
- Very-high-risk ASCVD: LDL-C goal generally  $<55$  mg/dL
- Other ASCVD: goal often  $<70$  mg/dL

### Triglycerides

- $\geq 500$  mg/dL: increased pancreatitis risk
- $\geq 1000$  mg/dL: high risk of pancreatitis—urgent management

## HYPERLIPIDEMIA IS HIGHLY MODIFIABLE



Heart-healthy (Mediterranean) diet



Regular aerobic activity



Weight management



Quit smoking



Control diabetes & blood pressure



Medications as indicated



Lower risk of heart attack and stroke



## RED FLAGS



TG  $>500$ – $1000$  mg/dL with epigastric pain → impending/active pancreatitis



Severe muscle pain, weakness, dark urine on statin → possible rhabdomyolysis



Chest pain, claudication, TIA → established ASCVD; needs urgent risk reduction



LDL-C  $\geq 190$  mg/dL or premature CAD family history → suspect FH; cascade screen relatives

01

IMAGE TITLE  
Hyperlipidemia Overview

CATEGORY  
Cardiology

DIFFICULTY  
Foundation

CLINICAL SOURCE  
2026 ACC/AHA Guideline on the Management of Dyslipidemia

REVIEW  
CLINICAL PEARLS

# HYPERLIPIDEMIA: LIPOPROTEIN ANATOMY & PHYSIOLOGY

Lipoproteins transport lipids through the blood. Their size, density, and apolipoprotein content determine their function and atherogenic potential.

## MAJOR LIPOPROTEIN CLASSES

- Apolipoprotein
- Triglyceride
- Cholesterol Ester
- Free Cholesterol
- Phospholipid

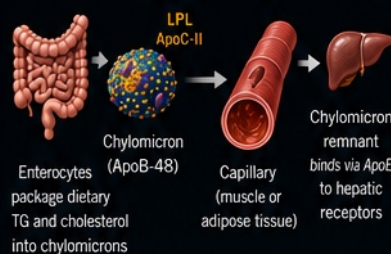
| CHYLOMICRON                          | VLDL                                | IDL                        | LDL                  | HDL                        |
|--------------------------------------|-------------------------------------|----------------------------|----------------------|----------------------------|
|                                      |                                     |                            |                      |                            |
| 75–1200 nm<br>ApoB-48, ApoC-II, ApoE | 30–80 nm<br>ApoB-100, ApoC-II, ApoE | 25–35 nm<br>ApoB-100, ApoE | 18–25 nm<br>ApoB-100 | 5–12 nm<br>ApoA-I, ApoA-II |

| DENSITY            | Least dense                            | Low density                      | Intermediate density               | Low density                               | Highest density                       |
|--------------------|----------------------------------------|----------------------------------|------------------------------------|-------------------------------------------|---------------------------------------|
| MAIN LIPID CONTENT | Triglycerides (dietary)                | Triglycerides (endogenous)       | Triglycerides & Cholesterol Esters | Cholesterol Esters                        | Protein & Phospholipids               |
| PRIMARY FUNCTION   | Deliver dietary TG to muscle & adipose | Deliver endogenous TG to tissues | TG remnants cleared by liver       | Deliver cholesterol to peripheral tissues | Remove cholesterol from tissues (RCT) |
| ATHEROGENICITY     | None                                   | Moderate                         | Moderate                           | High                                      | Protective                            |

★ Small, dense LDL → greatest arterial wall penetration and oxidation → highest atherogenic potential.

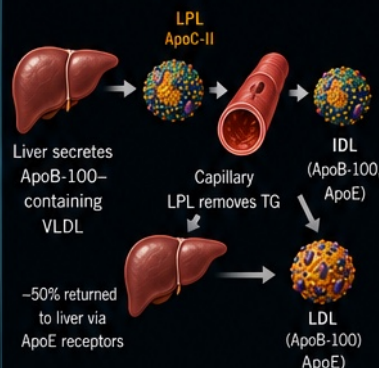
## LIPOPROTEIN METABOLISM

### 1 EXOGENOUS PATHWAY (DIETARY LIPIDS)

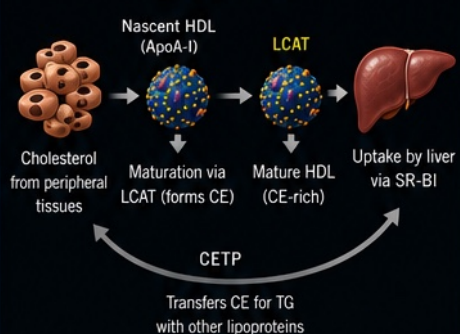


LPL hydrolyzes TG → Free fatty acids are taken up by tissues.

### 2 ENDOGENOUS PATHWAY (HEPATIC LIPIDS)



### 3 REVERSE CHOLESTEROL TRANSPORT (HDL-MEDIATED)



LPL = Lipoprotein Lipase LCAT = Lecithin-Cholesterol Acyltransferase  
ApoE = Ligand for hepatic remnant receptors CETP = Cholesteryl Ester Transfer Protein  
SR-BI = Scavenger Receptor Class B Type I

## KEY APOLIPOPROTEINS: STRUCTURE & FUNCTION

|  |          |                                | WHERE FOUND                                                                    |      |     |     |     |
|--|----------|--------------------------------|--------------------------------------------------------------------------------|------|-----|-----|-----|
|  |          |                                | CM                                                                             | VLDL | IDL | LDL | HDL |
|  | ApoB-48  | Chylomicrons                   | Structural protein; required for assembly and secretion of chylomicrons.       | ✓    | —   | —   | —   |
|  | ApoB-100 | VLDL, IDL, LDL                 | Structural protein; one ApoB-100 molecule per atherogenic particle.            | —    | ✓   | ✓   | —   |
|  | ApoA-I   | HDL                            | Activates LCAT; essential for HDL formation and reverse cholesterol transport. | ✓    | ✓   | ✓   | ✓   |
|  | ApoC-II  | Chylomicrons, VLDL, HDL        | Activates lipoprotein lipase (LPL) for triglyceride hydrolysis.                | ✓    | ✓   | —   | ✓   |
|  | ApoE     | Chylomicron remnants, IDL, HDL | Ligand for hepatic receptors (LRP, LDLR); clears remnants and IDL.             | ✓    | ✓   | ✓   | —   |

CM = Chylomicron VLDL = Very Low Density Lipoprotein IDL = Intermediate Density Lipoprotein  
LDL = Low Density Lipoprotein HDL = High Density Lipoprotein

## ★ CLINICAL PEARLS

**(B)** ApoB is the best indicator of atherogenic particle number. Target in high-risk patients (Goal: <90 mg/dL or ≥50% reduction).

**(Lp(a))** Lp(a) is genetically determined and independent of LDL-C. Elevated Lp(a) increases ASCVD risk.

**(Shield)** ApoB often predicts ASCVD risk better than LDL-C when discordant.

HDL quantity ≠ HDL quality. Function (cholesterol efflux) matters more than HDL-C number.

02

IMAGE TITLE  
Lipoprotein Anatomy  
& Physiology

CATEGORY  
Cardiology

DIFFICULTY  
Foundation

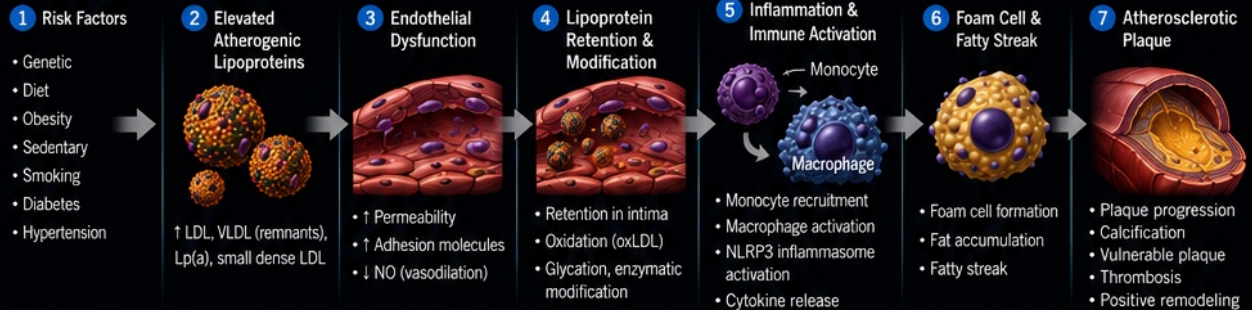
REFERENCE  
2026 ACC/AHA Guideline  
for the Management of Dyslipidemia

REVIEW  
CLINICAL  
PEARLS

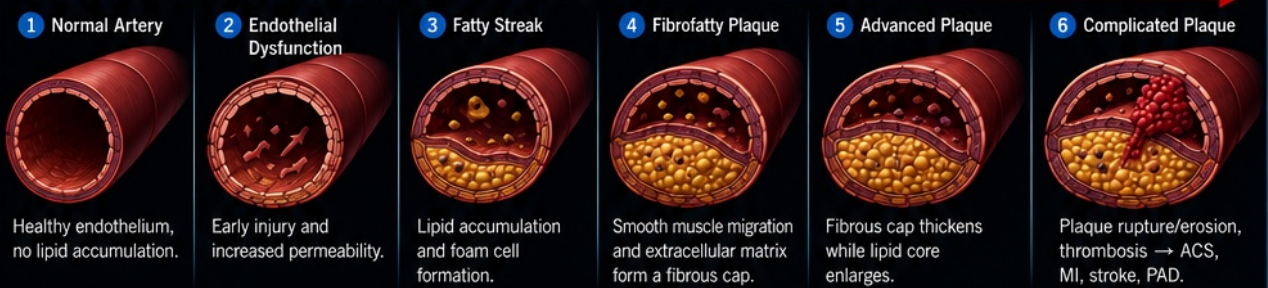
# HYPERLIPIDEMIA: PATHOPHYSIOLOGY

*Elevated atherogenic lipoproteins cause progressive arterial wall injury, lipid accumulation, inflammation, and plaque formation—driving atherosclerosis and ASCVD.*

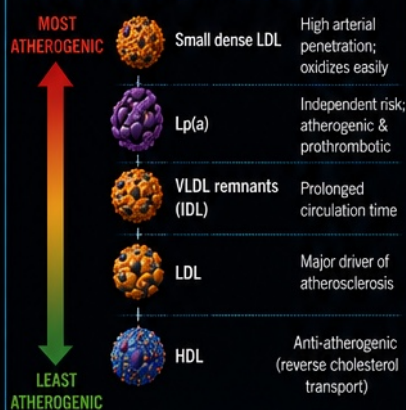
## THE PATHOGENIC CASCADE



## PROGRESSION OF ATHEROSCLEROSIS: YEARS TO DECADES



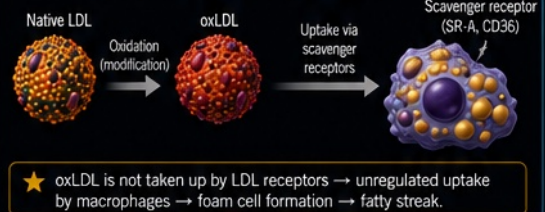
## LIPOPROTEINS & ATHEROGENICITY



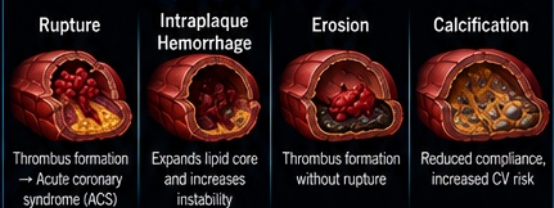
## ROLE OF Lp(a)



## OXIDIZED LDL → FOAM CELL



## PLAQUE COMPLICATIONS



## KEY MECHANISMS SUMMARY

- Endothelial dysfunction is the initiating event.
- LDL retention and modification drive inflammation.
- Chronic inflammation leads to plaque formation.
- Plaque instability causes thrombosis and clinical events.
- Lowering atherogenic lipoproteins reduces ASCVD risk.

## HIGH-YIELD PEARLS

- ★ The longer and higher the exposure to atherogenic lipoproteins, the greater the risk.
- Ⓑ Small dense LDL, Lp(a), and remnant cholesterol are especially atherogenic.
- 🛡️ Inflammation is central to every stage of atherosclerosis.
- 🎯 Atherosclerosis is a systemic disease, not just a cholesterol problem.
- 📊 ApoB concentration predicts lifetime exposure to atherogenic particles.
- 🔧 Aggressive risk factor modification stabilizes plaques and prevents events.

## CLINICAL CONSEQUENCES

- MI • Unstable Angina
- Sudden Cardiac Death
- Stroke • TIA
- PAD • Claudication
- Critical Limb Ischemia
- Peripheral Aneurysm (esp. abdominal)

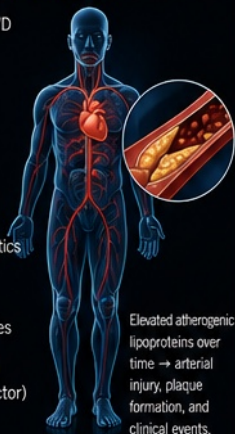
# HYPERLIPIDEMIA: CLINICAL PRESENTATION

*Most patients are asymptomatic until atherosclerotic cardiovascular disease (ASCVD) or complications develop. Physical findings are uncommon except in severe or inherited dyslipidemias.*

## PATIENT PROFILE & RISK FACTORS

- Family history of premature ASCVD
- Sedentary lifestyle
- Obesity, metabolic syndrome
- Hypertension, diabetes mellitus
- Smoking
- CKD, hypothyroidism, NAFLD
- Medications: steroids, retinoids, thiazides, beta blockers, antiretrovirals, atypical antipsychotics
- Diet high in saturated fat, ultra-processed foods, refined carbohydrates, and excess calories

★ **Elevated Lipoprotein(a) [Lp(a)]**  
(inherited, independent risk factor)



## TYPICAL CLINICAL PRESENTATION



Usually asymptomatic. Incidental abnormal lipid panel.



Symptoms are from ASCVD and may be the first presentation.

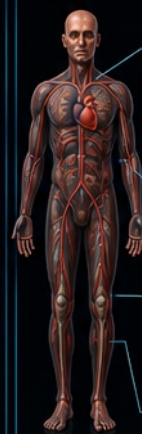


Severe hypertriglyceridemia (especially  $TG \geq 500$  mg/dL) → abdominal pain from acute pancreatitis.



Familial hypercholesterolemia may present with tendon xanthomas/xanthelasma.

## PRESENTATION OF ASCVD



### CORONARY ARTERY DISEASE

- Exertional chest pain
- Dyspnea
- Acute coronary syndrome
- Silent ischemia



### CEREBROVASCULAR DISEASE

- TIA
- Ischemic stroke
- Amaurosis fugax



### PERIPHERAL ARTERY DISEASE

- Claudication
- Decreased pulses
- Rest pain, nonhealing wounds



### AORTIC VALVE DISEASE

- Progressive calcific aortic stenosis
- Strongly associated with elevated Lp(a)

## PHYSICAL FINDINGS IN SEVERE OR GENETIC DYSLIPIDEMIAS

### XANTHELASMA (Common)



Yellow, flat or slightly elevated plaques on eyelids. Associated with  $\uparrow$  LDL or familial hypercholesterolemia.

### TENDINOUS XANTHOMAS (Classic)



Firm, nodular deposits over tendons (Achilles, extensors). Strongly associated with familial hypercholesterolemia.

### TUBEROUS XANTHOMAS (Severe)



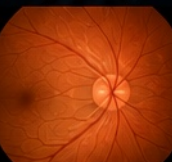
Firm nodules over pressure points (elbows, knees, buttocks). Occur in familial hypercholesterolemia (homozygous > heterozygous).

### XANTHOMA PLANARUM (Rare)



Yellow-orange flat plaques on palms, creases, or knees. Associated with familial dysbetalipoproteinemia (ApoE2/E2).

### LIPEMIA RETINALIS (Severe HTG)



Creamy, white discoloration of retinal vessels. Associated with severe hypertriglyceridemia ( $>1000$  mg/dL).

### ERUPTIVE XANTHOMAS (Severe HTG)



Yellow-red papules with erythematous halo, typically on extensor surfaces and buttocks. Associated with severe hypertriglyceridemia.

## COMPLICATIONS OF HYPERLIPIDEMIA



**Atherosclerotic Cardiovascular Disease (ASCVD)**  
MI, angina, stroke, TIA, PAD.



**Acute Pancreatitis (Severe HTG)**  
Risk markedly increased when  $TG >1000$  mg/dL.



**Aortic Stenosis (Calcific)**  
Associated with elevated LDL-C and Lp(a).



**Atherosclerotic Renal Disease**  
May lead to renovascular hypertension and CKD.



**Nonalcoholic Fatty Liver Disease (NAFLD)**  
Strongly linked to dyslipidemia and metabolic syndrome.

## RED FLAGS



Premature ASCVD (men  $<55$ ; women  $<65$ )



LDL-C  $\geq 190$  mg/dL (severe hypercholesterolemia)



$TG \geq 500$  mg/dL (severe HTG)  
 $TG \geq 1000$  mg/dL → very high pancreatitis risk



Xanthomas or corneal arcus before age 45



Premature or rapidly progressive ASCVD despite optimal therapy

## ASSOCIATED CONDITIONS



**Metabolic Syndrome / Insulin Resistance**  
Obesity, HTN, dysglycemia, low HDL, high TG.



**Diabetes Mellitus (Type 2)**

Common in atherogenic dyslipidemia.



**Hypothyroidism**

May cause secondary elevation in LDL-C.



**Chronic Kidney Disease**

Associated with dyslipidemia and  $\uparrow$  ASCVD risk.



**Familial Hypercholesterolemia (FH)**

Autosomal dominant; lifelong elevated LDL-C.



**Obstructive Sleep Apnea**

Increases CV risk and atherogenic lipids.



**Chronic Inflammatory Conditions / HIV**

Rheumatoid arthritis, psoriasis, lupus, HIV.

## ★ CLINICAL PEARLS



Most patients are asymptomatic until an ASCVD event occurs.



Look for physical stigmata in young patients or those with severe lipid abnormalities.



A detailed family history can reveal undiagnosed familial hypercholesterolemia.



Severe hypertriglyceridemia often presents first with pancreatitis, not ASCVD.



Lipid lowering reduces risk of MI, stroke, and cardiovascular death.



Identify and treat secondary causes of dyslipidemia.



Never miss FH: consider in anyone with LDL-C  $\geq 190$  mg/dL.

# HYPERLIPIDEMIA: PHYSICAL EXAMINATION

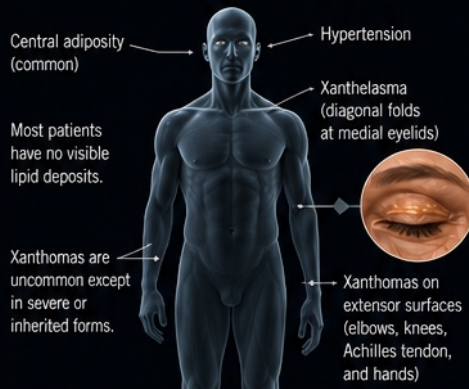
Physical findings of hyperlipidemia are uncommon except in severe or inherited forms.

Focus the exam on signs of atherosclerotic disease and associated metabolic conditions.

## EXAM OBJECTIVES

- ✓ Assess for stigmata of severe or genetic dyslipidemias
- ✓ Evaluate for complications of atherosclerotic disease
- ✓ Identify features of metabolic syndrome
- ✓ Identify secondary causes of dyslipidemia
- ✓ Risk-stratify and guide additional testing

## GENERAL APPEARANCE & VITAL SIGNS



## TARGETED CARDIOVASCULAR EXAM



## KEY HISTORY PEARLS

- 🕒 Premature ASCVD in first-degree relatives (men <55, women <65)
- 📋 Personal history of ASCVD or TIA/stroke
- 👉 Tendon xanthomas or xanthelasma in self or relatives
- 👉 Symptoms of hypothyroidism or nephrotic syndrome
- 💊 Medications that raise lipids
- 👉 Elevated Lp(a) in relatives or family history of early aortic stenosis

## PHYSICAL SIGNS OF SEVERE OR GENETIC DYSLIPIDEMIAS

### XANTHELASMA (Common)



Yellow, flat or slightly elevated plaques on medial eyelids. Associated with LDL elevation and familial hypercholesterolemia.

### TENDINOUS XANTHOMAS (Classic)



Firm, nodular deposits over tendons (Achilles, extensors). Strongly associated with familial hypercholesterolemia.

### TUBEROUS XANTHOMAS (Severe)



Firm nodules over pressure points (elbows, knees, buttocks). Occur in familial hypercholesterolemia (homozygous > heterozygous).

### XANTHOMA PLANARUM (Rare)



Yellow-orange flat plaques on palms, creases, or knees. Associated with familial dysbetalipoproteinemia (ApoE2 / E2).

### LIPEMIA RETINALIS (Severe HTG)



Creamy, white discoloration of retinal vessels. Associated with severe hypertriglyceridemia (>1000 mg/dL).

### ERUPTIVE XANTHOMAS (Severe HTG)



Yellow-red papules with erythematous halo, typically on extensor surfaces and buttocks. Associated with severe hypertriglyceridemia.

## EXAM FOR ATHEROSCLEROTIC DISEASE



### CAROTID ARTERIES

Auscultate for bruits; assess upstroke.



### CORONARY ARTERIES

Listen for murmurs (AS), assess for angina symptoms.



### PERIPHERAL ARTERIES

Check pulses, capillary refill, skin temperature, hair loss, ulcers, gangrene. Obtain ABI if symptomatic.



### ABDOMINAL AORTA

Palpate for enlargement or bruit.



### RENAL ARTERIES

Consider bruit in resistant hypertension.

## METABOLIC SYNDROME FAST CHECK

|                     | ABNORMAL IF                                 |
|---------------------|---------------------------------------------|
| Waist Circumference | >102 cm (40 in) men<br>>88 cm (35 in) women |
| Blood Pressure      | ≥130/85 mmHg                                |
| Fasting Glucose     | ≥100 mg/dL                                  |
| Triglycerides       | ≥150 mg/dL                                  |
| HDL Cholesterol     | <40 mg/dL men<br><50 mg/dL women            |

Diagnosis: ≥3 criteria present

## CLINICAL PEARLS FOR THE EXAM

- ★ Most patients have a normal exam.
- ★ Targeted exam should focus on:
  - ASCVD manifestations
  - Stigmata of severe/genetic dyslipidemias
  - Metabolic syndrome components
- ★ Tendon xanthomas before age 45 strongly suggest familial hypercholesterolemia.
- ★ Always examine peripheral pulses in patients with elevated LDL or multiple risk factors.
- ★ Do not forget to look for other causes of secondary dyslipidemia (thyroid, nephrotic syndrome, medications).
- ★ A normal physical examination does not exclude severe hyperlipidemia.

## RED FLAGS ON EXAM

- ⚠️ Tendon xanthomas
- ⚠️ Xanthelasma before age 40
- ⚠️ Xanthomas eruptive or planarum
- ⚠️ Corneal arcus before age 45
- ⚠️ Severe lipemia retinalis
- ⚠️ LDL-C ≥190 mg/dL



Evaluate for genetic or severe dyslipidemia

## WHEN TO SUSPECT GENETIC DYSLIPIDEMIA

**FAMILIAL HYPERCHOLESTEROLEMIA (FH)**  
LDL-C ≥190 mg/dL  
Personal or family history of premature ASCVD  
Tendon xanthomas  
Xanthelasma

**FAMILIAL COMBINED HYPERLIPIDEMIA (FCH)**  
↑ TG and/or LDL-C  
Family history of mixed dyslipidemia or ASCVD  
Often associated with metabolic syndrome

**FAMILIAL DYSBETALIPOPROTEINEMIA (Type III)**  
Palmar xanthomas  
Tuberous xanthomas  
Premature ASCVD  
TC and TG elevated

**FAMILIAL CHYLOMICRONEMIA SYNDROME**  
Recurrent pancreatitis  
Eruptive xanthomas  
Extreme HTG (>2000 mg/dL)  
Abdominal lipemia retinalis

**ELEVATED LIPOPROTEIN(a) (Lp(a))**  
Often no physical findings  
Strong family history  
Premature ASCVD  
Premature aortic stenosis

## ESSENTIAL NEXT STEPS

- ✓ Obtain fasting lipid panel
- ✓ Assess ASCVD risk
- ✓ Look for secondary causes
- ✓ Initiate lifestyle counseling
- ✓ Measure ApoB and obtain once-in-lifetime Lp(a)
- ✓ Early identification saves lives

# HYPERLIPIDEMIA: DIAGNOSTIC EVALUATION

Accurate diagnosis and risk stratification guide therapy intensity and prevent atherosclerotic cardiovascular disease (ASCVD).

## LIPID PANEL (FASTING 9–12 HOURS)

| PARAMETER                                         | DESIRABLE / OPTIMAL                                                              |
|---------------------------------------------------|----------------------------------------------------------------------------------|
| Total Cholesterol                                 | <200 mg/dL                                                                       |
| LDL-C (primary treatment target in most patients) | <100 mg/dL<br><70 mg/dL (very high risk)                                         |
| HDL-C                                             | ≥40 mg/dL (men)<br>≥50 mg/dL (women)                                             |
| Triglycerides                                     | <150 mg/dL                                                                       |
| Non-HDL-C                                         | <130 mg/dL<br><100 mg/dL (very high risk)                                        |
| ApoB                                              | <90 mg/dL (standard risk)<br><80 mg/dL (high risk)<br><65 mg/dL (very high risk) |



LDL-C reflects atherogenic cholesterol

Non-HDL-C captures all atherogenic particles

ApoB reflects the number of atherogenic particles

★ Repeat 4–12 weeks after therapy initiation or change, then every 3–12 months if stable.

## ADVANCED LIPID TESTING



### Lipoprotein(a) [Lp(a)]

Once-in-lifetime test.  
Elevated if >50 mg/dL (>125 nmol/L).  
Strong, independent risk factor.



### Apolipoprotein B (ApoB)

Optimal <90 mg/dL.  
Strong predictor of ASCVD risk.  
Reflects total atherogenic particles.



### LDL Particle Number (LDL-P)

Elevated if >1300 nmol/L.  
Useful in discordant cases.



### Remnant Cholesterol

Elevated in TG 150–499 mg/dL.  
Represents atherogenic remnant particles.

## SCREEN FOR SECONDARY CAUSES



Hypothyroidism



Uncontrolled diabetes mellitus



Chronic kidney disease



Cholestatic liver disease



Medications (e.g., steroids, retinoids, antiretrovirals, calcineurin inhibitors, thiazides, beta blockers, atypical antipsychotics)



Excess alcohol use



Pregnancy



Obstructive sleep apnea



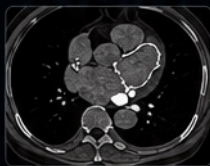
Nephrotic syndrome



Cushing syndrome

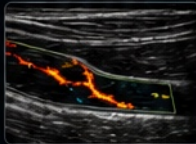
## ASSESS SUBCLINICAL ATHEROSCLEROSIS & TARGET ORGAN DAMAGE

### CORONARY ARTERY CALCIUM (CAC) SCORE



Non-contrast cardiac CT.  
CAC score: 0 = low risk.  
CAC ≥100 or ≥75th percentile = high risk.  
CAC = 0 may support deferring statin therapy in selected patients.

### CAROTID PLAQUE ULTRASOUND (Not CIMT)



Detects carotid plaque and stenosis.  
More predictive of ASCVD risk than CIMT.  
Not routinely recommended for risk stratification.

### ANKLE-BRACHIAL INDEX (ABI)



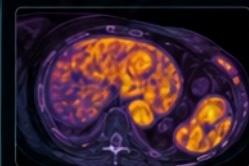
ABI <0.90 indicates peripheral artery disease.

### CORONARY CT ANGIOGRAPHY (CCTA)



Anatomic visualization of coronary plaque and stenosis in selected patients.

### ADVANCED IMAGING (WHEN INDICATED)



PET (inflammation), MRI (plaque characterization), or IVUS/OCT in high-risk or complex cases.

## FAMILIAL HYPERCHOLESTEROLEMIA (FH) EVALUATION



Personal or family history of premature ASCVD



LDL-C ≥190 mg/dL in adults



LDL-C ≥160 mg/dL in children/adolescents



Tendon xanthomas or corneal arcus <45 years



Use Dutch Lipid Clinic Network or Simon Broome criteria to assess probability of FH

### DUTCH LIPID CLINIC NETWORK CRITERIA (SIMPLIFIED)

| CRITERIA                                                       | POINTS |
|----------------------------------------------------------------|--------|
| LDL-C ≥330 mg/dL                                               | 8      |
| LDL-C 250–329 mg/dL                                            | 5      |
| LDL-C 190–249 mg/dL                                            | 8      |
| Tendon xanthomas                                               | 6      |
| Corneal arcus <45 yrs                                          | 4      |
| First-degree relative with premature ASCVD or LDL-C ≥190 mg/dL | 1      |
| DNA mutation (LDLR, APOB, PCSK9)                               | 8      |

Definite FH ≥8 | Probable FH 6–8 | Possible FH 3–5

## RISK ASSESSMENT & STRATIFICATION

Use PREVENT™ equations (or Pooled Cohort Equations where applicable) (lifetime course estimate)

### LOW RISK <5% 10-YEAR RISK

- Lifestyle modification
- Consider statin if risk enhancers present

### BORDERLINE RISK 5% – <7.5%

- Risk discussion
- Consider CAC score
- Consider moderate-intensity statin

### INTERMEDIATE RISK 7.5% – <20%

- Moderate- to high-intensity statin
- Goal LDL-C reduction ≥30%

### HIGH RISK ≥20% OR ASCVD PRESENT

- High-intensity statin (or max tolerated)
- Goal LDL-C reduction ≥50%
- Consider non-statin therapy

Risk enhancers: family history of premature ASCVD, Lp(a) >50 mg/dL, ApoB ≥130 mg/dL, CKD, metabolic syndrome, inflammatory diseases.

## SPECIAL CONSIDERATIONS



Severe Hypertriglyceridemia (≥500 mg/dL).  
Evaluate for pancreatitis risk. Check for secondary causes.  
Consider fibrate, omega-3 fatty acids.



Pregnancy  
Statins contraindicated. Consider bile acid sequestrants.  
In very high-risk FH, statins may be considered under specialist guidance.

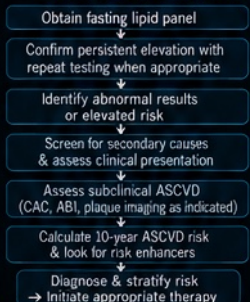


Older Adults (>75 years)  
Individualize therapy based on overall risk, comorbidities, and life expectancy.



Pediatric Patients  
Screen if family history of premature ASCVD or high LDL-C.  
Lifestyle first; consider statin ≥10 years in FH.

## DIAGNOSTIC ALGORITHM



A comprehensive evaluation ensures accurate diagnosis, identifies high-risk patients, and guides aggressive management to prevent ASCVD events.

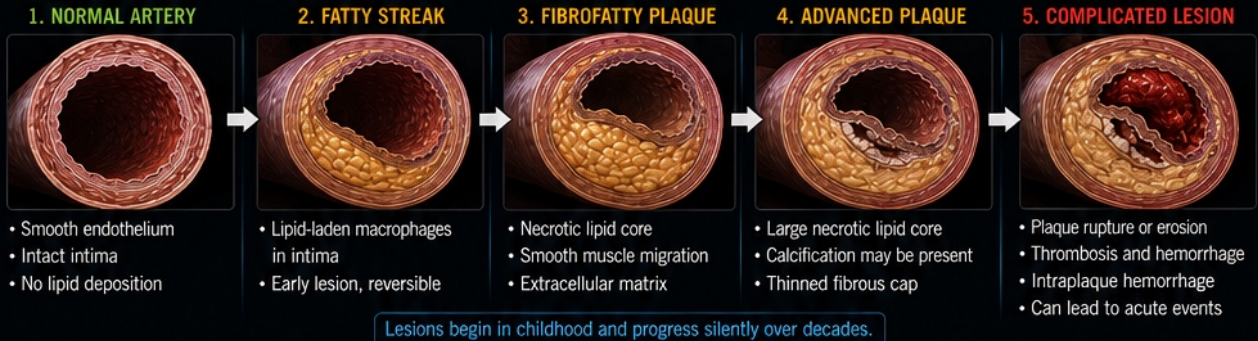
## KEY TAKEAWAYS

- ✓ Fasting lipid panel is the foundation.
- ✓ ApoB reflects total atherogenic particle burden.
- ✓ ApoB and Lp(a) provide additional risk information.
- ✓ Screen for secondary causes in all patients.
- ✓ Subclinical atherosclerosis testing refines risk assessment.
- ✓ FH should be suspected early and managed aggressively.
- ✓ Risk stratification determines treatment intensity.

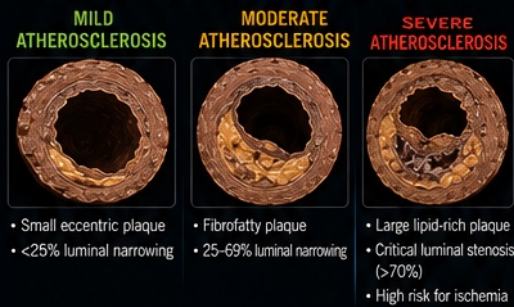
# HYPERLIPIDEMIA: GROSS PATHOLOGY

Chronic elevation of atherogenic lipids leads to progressive atherosclerosis and its complications.  
The following are key gross pathology findings in advanced disease.

## PROGRESSION OF ATHEROSCLEROSIS IN LARGE ELASTIC ARTERIES



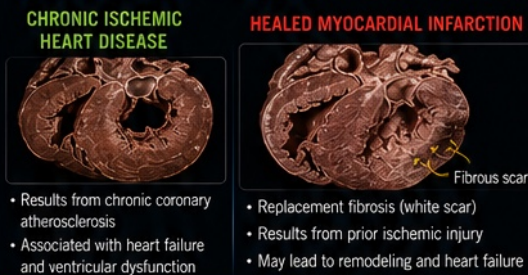
## CORONARY ARTERY: CROSS SECTIONS



## AORTA: GROSS SPECIMENS



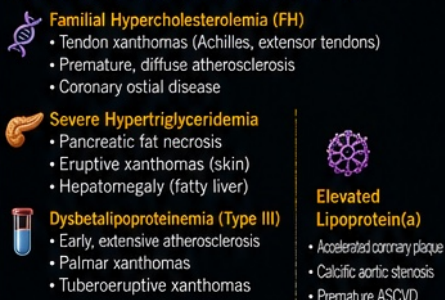
## HEART: GROSS PATHOLOGY



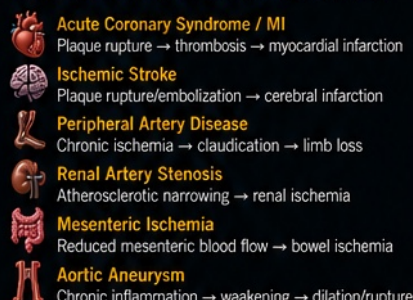
## PERIPHERAL ARTERY DISEASE (GROSS FINDINGS)



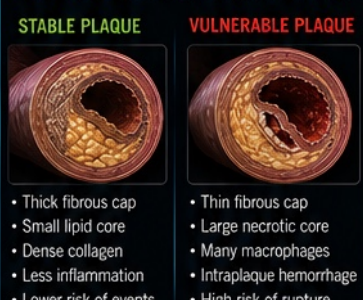
## GROSS FINDINGS BY LIPID DISORDER



## COMPLICATIONS OF ATHEROSCLEROSIS



## STABLE PLAQUE vs VULNERABLE PLAQUE



## PEARLS

- Atherosclerosis is a systemic, diffuse process.
- Plaques can be present even with normal or mildly elevated lipid levels.
- Plaque rupture, not stenosis severity, most often causes acute events.
- Aggressive lipid lowering can stabilize plaques and reduce events.
- Early detection and treatment prevent irreversible damage.
- Luminal stenosis often underestimates total plaque burden due to positive remodeling.

## Gross Pathology Key

- Early / Mild Disease
- Advanced Disease
- Established Disease
- Complicated / High Risk

# HYPERLIPIDEMIA: HISTOLOGY

Microscopic changes in atherosclerosis reflect progressive lipid accumulation, inflammation, smooth muscle response, and extracellular matrix remodeling.

## HISTOLOGIC STAGES OF ATHEROSCLEROSIS

| 1. NORMAL ARTERY                                                                                                                                                                                  | 2. FATTY STREAK                                                                                                                                                                            | 3. FIBROFATTY PLAQUE                                                                                                                                                                            | 4. ADVANCED PLAQUE                                                                                                                                              | 5. COMPLICATED LESION                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Intima<br/>Internal Elastic Lamina<br/>Media<br/>Adventitia</p> <ul style="list-style-type: none"> <li>Intact endothelium</li> <li>No lipid deposition</li> <li>Normal architecture</li> </ul> | <ul style="list-style-type: none"> <li>Foam-cell accumulation with early inflammatory activity</li> <li>Lipid-laden macrophages in intima</li> <li>No significant SMC migration</li> </ul> | <ul style="list-style-type: none"> <li>Lipid core formation</li> <li>Smooth muscle cell migration from media</li> <li>Extracellular matrix deposition</li> <li>Fibrous-cap formation</li> </ul> | <ul style="list-style-type: none"> <li>Fibrous cap overlying a necrotic lipid core</li> <li>Calcification may be present</li> <li>Neovascularization</li> </ul> | <ul style="list-style-type: none"> <li>Fibrous cap rupture or erosion</li> <li>Intraplaque hemorrhage</li> <li>Thrombosis</li> <li>Acute clinical events</li> </ul> |

Lesions begin in childhood and progress silently over decades.

## PLAQUE COMPOSITION AND FEATURES

- Fibrous Cap**: Collagen and smooth muscle cells; thickness determines plaque stability.
- Lipid Core**: Extracellular lipids, cholesterol clefts, and cell debris; prone to expansion and rupture.
- Inflammation**: Macrophages and T cells release cytokines and proteases.
- Calcification**: Microcalcification and confluent dystrophic calcification may occur.
- Cholesterol Clefts**: Needle-shaped spaces from dissolved cholesterol crystals within the necrotic core.
- Media**: Smooth muscle cells may proliferate and migrate into intima.
- Shoulder Region**: Macrophage-rich areas associated with rupture risk.

## STABLE PLAQUE vs VULNERABLE PLAQUE

| STABLE PLAQUE                                                                                                                                                                  | VULNERABLE PLAQUE                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Thick fibrous cap</li> <li>Small lipid core</li> <li>Dense collagen</li> <li>Less inflammation</li> <li>Lower risk of events</li> </ul> | <ul style="list-style-type: none"> <li>Thin fibrous cap—classically &lt;65 µm in coronary lesions</li> <li>Large necrotic core</li> <li>Many macrophages</li> <li>Intraplaque hemorrhage</li> <li>High risk of rupture</li> </ul> |

## CORONARY ARTERY: HISTOLOGY

| EARLY LESION                                                                                                                 | INTERMEDIATE LESION                                                                                                      | ADVANCED LESION                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Foam cells in intima</li> <li>GAG accumulation</li> <li>Intact endothelium</li> </ul> | <ul style="list-style-type: none"> <li>Fibrofatty plaque</li> <li>Smooth muscle cells</li> <li>ECM deposition</li> </ul> | <ul style="list-style-type: none"> <li>Necrotic core</li> <li>Fibrous cap</li> <li>Calcification possible</li> </ul> |

## AORTA: HISTOLOGY

| EARLY PLAQUE                                                                                          | ADVANCED PLAQUE                                                                                                           | COMPLICATED PLAQUE                                                                                             |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Subendothelial lipid deposition</li> <li>Foam cells</li> </ul> | <ul style="list-style-type: none"> <li>Dense fibrous tissue</li> <li>Cholesterol clefts</li> <li>Calcification</li> </ul> | <ul style="list-style-type: none"> <li>Ulceration</li> <li>Intraplaque hemorrhage</li> <li>Thrombus</li> </ul> |

## PERIPHERAL ARTERY: HISTOLOGY

| MILD DISEASE                                                                                                         | ADVANCED ATHEROSCLEROSIS                                                                                       | SEVERE PAD / NEAR-OCCLUSIVE PLAQUE                                                                                  |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Intimal thickening</li> <li>Minimal lipid</li> <li>Preserved lumen</li> </ul> | <ul style="list-style-type: none"> <li>Large lipid core</li> <li>Fibrosis</li> <li>Luminal stenosis</li> </ul> | <ul style="list-style-type: none"> <li>Extensive fibrosis</li> <li>Calcification</li> <li>Near-occlusion</li> </ul> |

## HISTOLOGIC FEATURES BY LIPID DISORDER

|                                                                                                                                                                                                               |                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Familial Hypercholesterolemia (FH)</b> <ul style="list-style-type: none"> <li>Accelerated fibroatheroma formation and premature plaque burden</li> <li>More extensive lipid core at earlier age</li> </ul> | <b>Elevated Lipoprotein(a)</b> <ul style="list-style-type: none"> <li>Accelerated coronary plaque</li> <li>Calcific aortic stenosis</li> <li>Premature ASCVD</li> </ul> |
| <b>Severe Hypertriglyceridemia</b> <ul style="list-style-type: none"> <li>Foam cells with neutral lipid</li> <li>Eruptive xanthomas (skin biopsy: foam cells in dermis)</li> </ul>                            |                                                                                                                                                                         |
| <b>Dysbetalipoproteinemia (Type III)</b> <ul style="list-style-type: none"> <li>Mixed remnant lipid deposition</li> <li>Cholesterol clefts</li> <li>Premature atherosclerosis</li> </ul>                      |                                                                                                                                                                         |

## COMPLICATIONS OF ATHEROSCLEROSIS

|                                                                                            |
|--------------------------------------------------------------------------------------------|
| <b>Acute Coronary Syndrome / MI</b><br>Plaque rupture or erosion → thrombosis → infarction |
| <b>Ischemic Stroke</b><br>Embolization or local thrombosis → cerebral infarction           |
| <b>Peripheral Artery Disease</b><br>Progressive stenosis → claudication → limb loss        |
| <b>Renal Artery Stenosis</b><br>Intimal hyperplasia → reduced renal perfusion              |
| <b>Mesenteric Ischemia</b><br>Plaque in mesenteric vessels → bowel ischemia                |
| <b>Aortic Aneurysm</b><br>Medial degeneration + inflammation → dilation/rupture            |

## PEARLS

- ★ Atherosclerosis is a systemic, diffuse process.
- ★ Luminal stenosis often underestimates total plaque burden due to positive remodeling.
- ★ Plaques can be present even with normal or mildly elevated lipid levels.
- ★ Plaque rupture, not stenosis severity, most often causes acute events.
- ★ Aggressive lipid lowering can stabilize plaques and reduce events.
- ★ Early detection and treatment prevent irreversible damage.

# HYPERLIPIDEMIA: TREATMENT & MANAGEMENT

*Atherosclerosis is progressive but treatable.*

*Lower LDL-C early, lower it aggressively, and sustain it lifelong.*

## LIPID MANAGEMENT PYRAMID



## LIPID LOWERING THERAPIES

| THERAPY                                   | LDL-C REDUCTION | OUTCOME BENEFIT                   | KEY POINTS                                                    |
|-------------------------------------------|-----------------|-----------------------------------|---------------------------------------------------------------|
| <b>High-intensity Statin</b>              | ≈50–60%         | ↓ ASCVD events by ≥25%            | First-line therapy for most patients                          |
| <b>Ezetimibe</b>                          | ≈18–25%         | ↓ ASCVD events (add-on to statin) | Well tolerated; oral once daily                               |
| <b>PCSK9 mAb</b> (Alirocumab/ Evolocumab) | ≈50–60%         | ↓ ASCVD events (add-on to statin) | Subcutaneous q2–4 weeks                                       |
| <b>Bempedoic Acid</b>                     | ≈17–20%         | ↓ ASCVD events (CLEAR Outcomes)   | Oral; option if statin intolerant                             |
| <b>Inclisiran</b>                         | ≈50%            | Outcome benefit emerging          | Subcutaneous at baseline, 3 months, then every 6 months       |
| <b>Icosapent Ethyl</b> (High TG)          | ≈ -             | ↓ CV events (REDUCE-IT)           | For selected patients with TG 135–499 mg/dL on statin therapy |

Percent reductions are approximate and vary by baseline LDL-C and patient factors.

## TREATMENT ALGORITHM

### RISK-BASED, SHARED DECISION-MAKING

- Confirm lipid abnormality and evaluate for secondary causes of dyslipidemia
- Assess overall risk using PREVENT-ASCVD risk estimator or identify high-risk condition (ASCVD, diabetes, CKD, FH, etc.)
- Define LDL-C/non-HDL-C goal and required percentage reduction based on risk
- Initiate maximally tolerated statin when indicated; consider early combination therapy if needed to reach goal
- Reassess in 4–12 weeks (use fasting lipid panel when clinically indicated) and intensify therapy as needed
- Reinforce lifestyle therapy and adherence; monitor for efficacy and safety long term

## MONITORING & FOLLOW-UP

- Repeat lipid panel 4–12 weeks after therapy initiation or change, then every 3–12 months thereafter
- Assess adherence, tolerability, and lifestyle factors
- Monitor for statin adverse effects (CK, liver enzymes if clinically indicated)
- Reassess ASCVD risk periodically and adjust treatment intensity as needed

## SPECIAL POPULATIONS

- Familial Hypercholesterolemia (FH)**  
Treat aggressively; consider combination therapy early. Cascade screen family members.
- Older Adults (>75 years)**  
Individualize therapy based on life expectancy, comorbidities, and patient preference.
- Diabetes Mellitus**  
Statin therapy recommended for adults 40–75 with diabetes. Consider earlier in high-risk patients.
- Chronic Kidney Disease**  
Use guideline-directed statin therapy with attention to eGFR and drug interactions. Avoid or dose-adjust fibrates in advanced CKD.
- Pregnancy**  
Discontinue statins in most patients. Individualized continuation may be considered for exceptionally high-risk patients (e.g., homozygous FH, prior ASCVD) under specialist guidance. Bile-acid sequestrants may be used when appropriate.
- Statin Intolerance**  
Use lowest tolerated dose, alternate-day dosing, or non-statin therapies. Referral to lipid specialist may help.

## LIFESTYLE THERAPY (FOUNDATION OF CARE)

- Diet:** Mediterranean-style or DASH diet. Emphasize vegetables, fruits, whole grains, legumes, nuts, and fish; limit saturated & trans fats and refined carbohydrates.
- Weight Management:** For patients with excess adiposity, aim for sustainable weight reduction—often 5–10% initially.
- Exercise:** ≥150 minutes/week of moderate-intensity aerobic activity + resistance training 2 days/week.
- Tobacco:** Complete cessation of all tobacco products and nicotine.
- Alcohol:** Limit to ≤1 drink/day (women) or ≤2 drinks/day (men). Avoid alcohol when triglycerides are severely elevated.
- Sleep & Stress:** 7–9 hours/night; manage stress to support cardiovascular health.

## LDL-C & NON-HDL-C TARGETS (2026 ACC/AHA GUIDELINE)

| CLINICAL CATEGORY                                                        | LDL-C GOAL                                | NON-HDL-C GOAL | APO B GOAL (ADJUNCTIVE) |
|--------------------------------------------------------------------------|-------------------------------------------|----------------|-------------------------|
| Primary Prevention (PREVENT risk-based)                                  | Risk-based, shared decision-making        | Risk-based     | Risk-based              |
| High-Risk Primary Prevention (e.g., diabetes, CKD 3–4, ≥7.5% 10-yr risk) | Often <70 mg/dL                           | <100 mg/dL     | <80 mg/dL               |
| ASCVD (Not Very High Risk)                                               | <70 mg/dL                                 | <100 mg/dL     | <80 mg/dL               |
| Very High-Risk ASCVD                                                     | <55 mg/dL                                 | <85 mg/dL      | <65 mg/dL               |
| Severe Hypercholesterolemia / FH                                         | ≥50% reduction + individualized threshold | <100 mg/dL     | <80 mg/dL               |

Targets are approximate. Individualize based on patient factors, comorbidities, and discussion of risks and benefits.

## MANAGEMENT OF SEVERE HYPERTRIGLYCERIDEMIA

- TG ≥500 mg/dL:** Address secondary causes, optimize lifestyle (low refined carbohydrates), limit alcohol. Assess ASCVD and pancreatitis risk.
- TG ≥1000 mg/dL:** Institute very-low-fat dietary intervention, treat secondary causes, and consider fibrate and/or prescription omega-3 therapy to reduce pancreatitis risk.

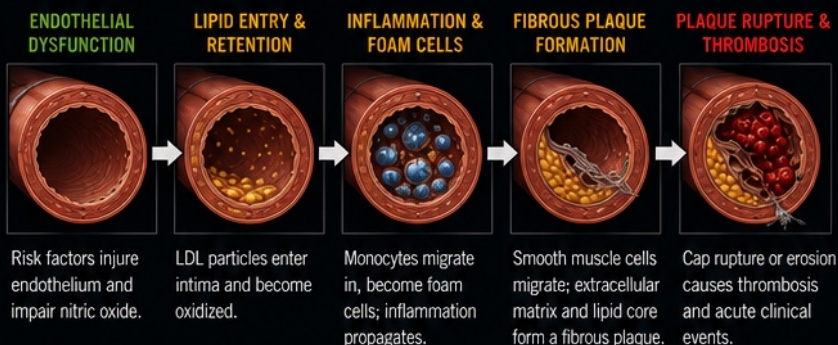
## MANAGEMENT PEARLS

- ★ Lower is better, and lower earlier is better.
- ★ LDL-C is the primary target in most patients.
- ★ ApoB reflects total atherogenic particle burden.
- ★ Combination therapy is often needed to achieve goals.
- ★ Plaque stabilizes and can regress with intensive lipid lowering.
- ★ Adherence to therapy and lifestyle determines long-term success.

# HYPERLIPIDEMIA: CLINICAL PEARLS

*Lower LDL-C early, lower it to the appropriate risk-based goal, and sustain it lifelong to prevent atherosclerotic events.*

## THE ATHEROSCLEROTIC PROCESS



A dynamic, chronic disease that begins early and progresses silently over decades.

## TREAT TO TARGET

Achieve and maintain LDL-C goals based on risk and required reduction.

Lower is better, and lower earlier is better.

Intensity matters. Higher initial intensity achieves goals sooner.

Sustain it. Long-term adherence preserves benefit.

Reassess and adjust. Therapy should be dynamic and individualized.

## PRIMARY PREVENTION GOALS

Delay or prevent the first ASCVD event.

- Address lifetime risk starting early in adulthood.
- Emphasize lifestyle, risk factor control, and LDL-C reduction.
- Use non-HDL-C when triglycerides are elevated; ApoB can identify residual atherogenic-particle risk, particularly with TG >200 mg/dL, diabetes, or very low achieved LDL-C.
- Earlier intervention yields a longer period of event-free life.



## SECONDARY PREVENTION GOALS

Prevent recurrent events and progression.

- Aggressive risk factor modification is essential.
- Achieve lower LDL-C goals (<55 mg/dL for very-high-risk ASCVD).
- Combine therapies when needed.
- Every 1 mmol/L (~39 mg/dL) drop in LDL-C reduces major vascular-event risk by roughly 20–25%.



## RISK ENHANCERS & MODIFIERS

- Family history of premature ASCVD (men <55, women <65)
- Elevated Lipoprotein(a) ≥50 mg/dL (≥125 nmol/L)
- Metabolic syndrome
- Chronic kidney disease (eGFR 15–59 mL/min/1.73 m<sup>2</sup>)
- Chronic inflammatory conditions (RA, psoriasis, HIV, etc.)
- Smoking
- Persistently elevated triglycerides (≥175 mg/dL)
- ApoB ≥130 mg/dL
- South Asian ancestry
- Subclinical atherosclerosis (CAC burden and percentile refine risk; CAC ≥100 strongly favors treatment, ABI <0.90, carotid plaque)

## CLINICAL PEARLS

- ★ Atherosclerosis is a systemic disease.
- ★ LDL-C is the primary target for atherosclerotic risk reduction.
- ★ Time and exposure matter: cumulative LDL-C drives risk.
- ★ Combination therapy is often required to achieve and sustain goals.
- ★ Intensive lipid lowering stabilizes plaque and may produce modest regression.
- ★ Lifestyle therapy amplifies every medication.
- ★ Patient education and adherence are as important as the prescription.
- ★ Use shared decision-making and personalize therapy.
- ★ Start early. Benefit lasts a lifetime.

## LIFESTYLE: EVERY PERCENT COUNTS

- Heart-healthy diet:** Mediterranean or DASH pattern. Emphasize vegetables, fruits, whole grains, legumes, nuts, fish, and olive oil. Limit saturated fat, trans fat, refined carbs, and red/processed meats.
- Physical activity:** ≥150 min/week of moderate-intensity aerobic activity + resistance training 2 days/week.
- Weight management:** Even 5–10% weight loss improves lipids and metabolic health.
- Tobacco:** Complete cessation.
- Alcohol:** Limit (women ≤1 drink/day, men ≤2 drinks/day). Avoid when TG are high.

## MEDICATIONS: USE WISELY

- When lipid-lowering medication is indicated, a maximally tolerated statin is the foundation for most patients.
- Add ezetimibe if goals not met.
- Add PCSK9 inhibitor, bempedoic acid, or inclisiran when needed.
- Icosapent ethyl (EPA) for selected high-risk patients with elevated TG.
- Consider safety, cost, access, and patient preference.

## THE LONG-TERM COMMITMENT



Sustained lipid control is a lifelong journey, not a short-term goal.

Small steps today lead to a lifetime of protection.

## DON'T FORGET

- Rule out secondary causes.
- Use fasting lipids when indicated.
- Measure Lp(a) at least once in adulthood.
- Calculate risk and set appropriate goals.
- Engage the patient to set expectations.
- Monitor, reinforce, and intensify as needed.
- Prevention saved today is quality life tomorrow.

“ The best time to lower LDL-C was yesterday. The second best time is today. — Rounds Codex Pearl ”